A Theory of Economic Slack

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Draft version: July 2026
Draft URL: pascalmichaillat.org/18/

Preface

This book is based for the most part on research that I conducted during graduate school at the University of California–Berkeley under the supervision of George Akerlof and Yuriy Gorodnichenko (2006–2010), and for fifteen years after that in collaboration with Emmanuel Saez (2010–2025).

In this preface, I explain what motivated the research, and provide the key references on which the book is based. The purpose is not to describe results, but to explain how the theory presented in the book emerged from a sequence of research questions, themselves motivated by macroeconomic puzzles and policy debates, and how it built on previous advances. Readers who prefer to begin with the case for putting slack back at the center of business cycle research may skip ahead to chapter 1.

Why do job seekers queue in recessions?

Early on in graduate school I read about Okun's misery index—the sum of the unemployment rate and inflation rate, designed as a simple measure of welfare in the US economy.[1] A low index means that the economy is doing well; a high index that it is doing poorly. The misery index posits that inflation and unemployment are the two main diseases affecting developed economies. The discovery of the misery index prompted me to start working on inflation. I particularly wanted to understand why seemingly tiny changes in the price of a baguette in France outraged customers so much (not least my mother). But that project proved too challenging, so I turned to the other component of the misery index: unemployment.

A first book that George advised me to read to start thinking about unemployment and business cycles was *Prices & Quantities* by Arthur Okun (1981). In it, Okun sketches a new macroeconomic framework in which labor and product markets are organized around implicit contracts between sellers and buyers. Although the slackish business cycle model

[1] See Asher, Defina, and Thanawala (1993) for a history of the misery index.

developed in the book is quite different from the framework that Okun had in mind, the book draws in numerous places on evidence that Okun collected, observations that he made, and references that he provided. Generally, the book shares Okun's conviction that Walrasian theory is inappropriate to describe labor and product markets, and that prices and wages are determined by norms that are somewhat insulated from cyclical forces.

George also suggested that I read *Equilibrium Unemployment Theory* by Christopher Pissarides (2000)—the leading textbook on unemployment theory. The textbook introduces the Diamond-Mortensen-Pissarides (DMP) model of the labor market, which was developed by Pissarides together with Peter Diamond and Dale Mortensen in the 1970s and 1980s. For their work, they received the Nobel Prize in Economics in 2010.[2]

An idea in the textbook that stuck with me is that it is possible, and in fact quite fruitful, to summarize the complex matching process in the labor market by a matching function—without modeling the matching process explicitly.[3] The idea plays a central role in this book, as trade in all markets is mediated by a well-behaved matching function. The function gives the number of trades that actually take place when a certain number of buyers and sellers are looking for each other on a market.

Reading the textbook, however, I was struck that the DMP model does not allow for the possibility—which seemed relevant in real life—that sometimes the economy just does not have enough jobs for all workers. This is an old, very Keynesian idea. Keynes had seen what was happening at the time of the Great Depression; he had seen the queues of workers waiting to apply for jobs, or to be hired, in front of job bureaus and factory gates. The photos collected in this preface, taken across the United States by Dorothea Lange, illustrate what Keynes (1936) had seen and wrote about in the *General Theory*: that in bad times there are just not enough jobs for everyone at the going wage, that jobs are rationed.

By ignoring job rationing, I felt, the DMP model cannot provide a good description of what happens in the labor market. In bad times, in the real world, workers queue for jobs. In that situation, a factory manager can just pick any worker that they want immediately; it is costless for firms to hire; matching becomes irrelevant. And yet, workers visibly queue for jobs. So it seems that something else happens in this situation: jobs are lacking.[4]

Thus, I became convinced that to describe the labor market better, we must allow for job rationing. My PhD thesis therefore developed a labor market model that features not only frictional unemployment—as in the DMP model—but also rationing unemployment—

[2] Albrecht (2011) reviews the early work by Diamond, Mortensen, and Pissarides, and the literature that built on it. Diamond (2011), Mortensen (2011), and Pissarides (2011) provide their own perspective on their research and the literature in their Nobel lectures.

[3] This idea goes back to work by Pissarides (1979, 1984, 1985, 1986). Before that, people usually modeled specific trading processes that resulted in trading probabilities between 0 and 1, often from the urn-ball model that is commonly used in probability.

[4] In the 1970s and 1980s, the literature on efficiency wages built models producing job rationing. But the models fell out of favor, maybe because they could not easily be integrated into the fast-growing DMP framework. For surveys of the efficiency-wage literature, see Akerlof (1984), Yellen (1984), and Katz (1986).

A. Howard Street, known as "Skid Row," the district of the unemployed (San Francisco, CA, 1937)

B. Unemployed men sitting on the sunny side of the San Francisco Public Library (San Francisco, CA, 1937)

C. Part of the daily lineup outside the State Employment Service Office (Memphis, TN, 1938)

D. Typical scene reflecting large population of unemployed in desperate need of work and looking for jobs (Camden, NJ, 1935)

Job rationing during the Great Depression in the United States

Sources. A: Dorothea Lange, available at `https://www.loc.gov/pictures/item/2017769707/`. B: Dorothea Lange, available at `https://www.loc.gov/pictures/item/2017769687/`. C: Dorothea Lange, available at `https://lccn.loc.gov/2017770574`. D: Franklin D. Roosevelt Library Public Domain Photographs, available at `https://catalog.archives.gov/id/195658`.

as in the *General Theory*. Technically, the issue is that the DMP model's labor demand is degenerate: it pins down a fixed level of labor market tightness, irrespective of the level of employment, instead of a proper downward-sloping demand for labor. I resolved this by replacing the standard linear production function, which has constant returns to labor, with a concave production function, which has diminishing returns to labor. This simple change delivers a downward-sloping labor demand curve, and with it, job rationing. In the model, firms may not want to create enough jobs for all the workers who want to work. This core idea was then published in the *American Economic Review* in 2012 (Michaillat 2012), and a streamlined version of the AER labor market model is presented in chapter 10.

Wages posed a related challenge. Shimer (2005) showed that with wages set by Nash bargaining, wages are too flexible: they absorb almost all of a productivity shock, leaving too little of it to generate sizable fluctuations in unemployment. Resolving this puzzle turned out to be straightforward, because Hall (2005) had already shown that a rigid wage—one that does not fully track productivity—remains compatible with bilateral efficiency in the DMP model, and generates realistic labor market fluctuations. Blanchard and Gali (2010) went further, showing that the wage need not be completely fixed: it can instead be a rigid function of productivity and still generate realistic fluctuations. With a rigid wage, unemployment fluctuated, and a surprising result appeared: the recessionary tide does not lift all boats. By that I mean that in recessions, when unemployment goes up, rationing unemployment also goes up—jobs are scarcer—but frictional unemployment goes down.

An important idea in the paper—one that carries through to the book—is that the slackish model readily accommodates realistic, somewhat rigid price and wage norms, and that such rigidity generates realistic business cycle fluctuations. But how does one build realistic price and wage norms? In the real world, almost no prices are set at auction, as the Walrasian model assumes; instead, most sellers and buyers are engaged in long-term relationships in which norms of fairness matter a great deal. In a paper with Erik Eyster and Kristof Madarasz that stemmed from the baguette project, we documented these norms of fairness in the product market and modeled pricing under fairness concerns (Eyster, Madarasz, and Michaillat 2021). Earlier, Akerlof (1980, 1982) and Akerlof and Yellen (1990) had documented these norms of fairness in the labor market and modeled wage-setting under fairness concerns. The book does not use the pricing and wage-setting models developed in those papers, but it draws on many of the facts documented there to build realistic price and wage norms, especially in chapters 5 and 6.

Towards the end of my PhD thesis, I stumbled upon an interesting property of the model: it is highly state dependent, in that it operates very differently in booms and slumps. As a result, some policies are more effective when the economy is slack, while others are more effective when the economy is tight. For instance, the fiscal multiplier is larger when unemployment is high, thus explaining Yuriy and Alan Auerbach's discovery that fiscal multipliers are higher during recessions (Auerbach and Gorodnichenko 2012). This result was published in the *American Economic Journal: Macroeconomics* in 2014 (Michaillat 2014). The state dependence is present throughout the book in various contexts, but especially in chapters 11 and 13, where we obtain slack-dependent policy multipliers.

Should unemployment insurance be extended in recessions?

I went on the job market in 2009–2010, in the aftermath of the Great Recession. One of the big policy questions at the time was whether it was warranted to increase the generosity of unemployment insurance. Emmanuel attended a mock job talk that I gave in December 2009, thought that the model that I had developed in my thesis could be helpful to think about the question, and reached out to see if I wanted to collaborate with him and Camille Landais, who was doing a postdoc at Berkeley at the time, on a project to determine how unemployment insurance should be adjusted over the business cycle.

The model was promising because public economics could only take into account supply-side factors—how unemployment insurance affects job search—but did not have the tools to include in the analysis demand-side factors—the number of workers that firms want to hire. The key idea was that if firms did not want to hire more workers, then searching more for jobs would only result in a giant rat race, but not additional employment, which would severely reduce the purported welfare cost of higher unemployment insurance and make extending unemployment insurance in bad times more desirable.

So in the summer of 2010, Camille, Emmanuel, and I started working on the following question: how should the generosity of unemployment insurance respond to unemployment fluctuations? We found that in the presence of job rationing, optimal unemployment insurance is more generous than the conventional Baily (1978)-Chetty (2006) level when the labor market is inefficiently slack and less generous when the labor market is inefficiently tight. The implication was that unemployment insurance should become more generous in bad times—as it is in practice in the United States. The project was ultimately published in 2018, in a pair of papers in the *American Economic Journal: Economic Policy* (Landais, Michaillat, and Saez 2018b,a). Unemployment insurance is discussed in chapter 11.

The AEJ papers aimed to express the formula for optimal unemployment insurance in sufficient statistics. The idea was to express optimal policy in terms of statistics that apply across a range of models and that can be directly measured.[5] Sufficient statistics make the policy tradeoffs transparent and rooted in empirical evidence, but they had not been used to study the market-level effects of policy—those were typically studied in clunky structural models that kept the policy tradeoffs and mechanisms obscure. The AEJ papers

[5] See Chetty (2009) and Kleven (2021) for surveys of the sufficient-statistic approach. Emmanuel's job-market paper played a central role in the shift towards sufficient statistics in public economics (Saez 2001).

brought sufficient statistics to the macroeconomic design of policy; the book follows this methodology, especially in part V.

How are aggregate demand and labor demand connected?

After our project on unemployment insurance, Emmanuel and I decided to keep working together with the framework. A natural question that arose is why labor demand is sometimes high and sometimes low. In the model from my thesis, which we then used in our unemployment insurance project, fluctuations in labor productivity lead to fluctuations in labor demand. This is the same as in the DMP model. But in reality, labor productivity is unlikely to be the main driver of business cycles. Surely fluctuations in aggregate demand, stemming from changes in monetary policy or animal spirits, influence labor demand. The question was therefore how to connect aggregate demand to labor demand.

The answer, it turned out, was slack in the product market. Unemployment is the most studied form of slack but slack also exists in the product market: firms are always looking for customers to buy their products and services.[6] Using the architecture developed by Barro and Grossman (1971), we built a model in which both product and labor markets are organized around matching functions and therefore display slack. In the model, when prices and wages are rigid, aggregate demand influences unemployment. For example, after an increase in aggregate demand, firms find more customers for their products, which reduces the idle time of their employees and makes labor more profitable. This increases firms' labor demand, which reduces unemployment.

The model therefore features frictional unemployment as well as classical unemployment, caused by excessive real wages, and Keynesian unemployment, caused by insufficient product demand—thus bridging the DMP and Old Keynesian theories of unemployment.[7] This work was published in the *Quarterly Journal of Economics* in 2015 (Michaillat and Saez 2015) and is covered in chapter 15.

While our QJE paper offers a simple model that is useful to understand why slack exists and how slack on different markets interact, it is static so cannot be used to think about interest rates, which inherently involve dynamic saving decisions. And interest

[6] Early contributions to the literature on product market slack include Diamond (1982, 1984), Walsh (1986), and Diamond and Yellin (1990). Diamond and Fudenberg (1989) and Boldrin, Kiyotaki, and Wright (1993) find that the model with product market slack exhibits periodic orbits; thus, models with slack might produce endogenous business cycles. Around the time of the Great Recession, several research teams started developing new models with product market slack, maybe inspired by Hall (2007), who had applied the DMP model to the product market to explain price rigidity and generate macroeconomic fluctuations. This crop of models includes Arseneau and Chugh (2007), Lehmann and Van der Linden (2010), Matha and Pierrard (2011), Gourio and Rudanko (2014), Petrosky-Nadeau and Wasmer (2015), and Bai, Rios-Rull, and Storesletten (2026).

[7] See Benassy (1993) for an overview of the Old Keynesian model. The model was initially known as the General Disequilibrium model (Barro and Grossman 1971), but it is now referred to as the Old Keynesian model (Barro 2025)—using the same terminology as Tobin (1992, 1993) to contrast with the New Keynesian model.

rates are the tool that central banks use to stabilize the economy. So, Emmanuel and I next developed a dynamic version of the model with interest rates. In this dynamic model, monetary policy sets interest rates, which determine aggregate demand, and through it unemployment. Hence, monetary policy can be used to stabilize the economy. This work was published in the *Oxford Economic Papers* in 2022 (Michaillat and Saez 2022). The model is covered in chapter 13.

What happens during long zero-lower-bound episodes?

One of the challenges that we faced in developing the OEP paper was how to build a nondegenerate aggregate demand curve. By nondegenerate I mean an aggregate demand curve that slopes downward, just like the old IS curve, and that can be used to study a monetary model at the zero lower bound or away from it. The standard aggregate demand curve, arising from a standard Euler equation, fails these requirements, because it is horizontal: it only tolerates a real interest rate equal to the time discount factor.

We discovered that it was possible to obtain such an aggregate demand curve by assuming that people's wealth directly enters their utility function. The justification for the assumption is that wealth is a marker of social status, and people value high social status. The assumption of wealth in the utility function produces a well-behaved aggregate demand curve, so we use it throughout part IV. The assumption had been used sporadically in macroeconomics after Kurz (1968) had first introduced it, especially to understand saving and investment behavior, and relatedly, long-run growth.[8] But it had not been used in business cycle research, so we could not publish the OEP paper until 2022, although it was written in 2014.

What we did manage to publish was a related paper showing that the wealth-in-the-utility assumption resolved all the anomalies of the New Keynesian model at the zero lower bound. That paper appeared in the *Review of Economics and Statistics* in 2021 (Michaillat and Saez 2021b). In chapter 13 we use the approach developed in the REStat paper to analyze forward guidance in the slackish business cycle model.

How large should stimulus packages be?

When monetary policy is constrained by the zero lower bound—as during the Great Depression and the Great Recession—the Federal Reserve is not able to stabilize the unemployment rate to its efficient level. In that case, monetary policy is ineffective, but fiscal policy can help stabilize unemployment. This is why the US government implemented a large stimulus package during the Great Recession. But it was unclear at the time how

[8] See for instance Zou (1994), Bakshi and Chen (1996), Carroll (2000), Piketty (2011), and Kumhof, Ranciere, and Winant (2015).

large that stimulus package should be, or even if a stimulus package was desirable at all, because there was no framework to answer the question.

The policy debate motivated Emmanuel and me to study optimal stimulus spending when there is unemployment, and in particular unemployment is inefficiently high. Public economics studied the optimal provision of public goods in normal times, but not in bad times (Samuelson 1954, 1955). Macroeconomics focused on the size of fiscal multipliers in bad times, but not on the optimal amount of public spending in bad times. Using a sufficient-statistic approach again, we characterized how optimal public spending deviates from the Samuelson rule when the unemployment gap is positive. We found that the size of the optimal stimulus package depends on several sufficient statistics, including the unemployment gap and the fiscal multiplier. These results were published in the *Review of Economic Studies* in 2019 (Michaillat and Saez 2019), and are covered in chapter 17.

How wide is the unemployment gap?

The AEJ and REStud papers showed that optimal unemployment insurance and optimal public spending depended critically on the unemployment gap. The results were not easily applicable, however, because we did not know what the efficient unemployment rate, u^*, was, so we did not know what the unemployment gap, $u - u^*$, was.

So our next project was to estimate the efficient unemployment rate and unemployment gap. We found that the efficient unemployment rate depended on a few sufficient statistics, which we measured in the United States, to find that the US unemployment gap is countercyclical—low (even negative) in good times and high (sharply positive) in bad times. This finding confirmed that unemployment insurance and public spending should be adjusted over the business cycle. This project was published in the *Journal of Public Economics Plus* in 2021 (Michaillat and Saez 2021a). The results are covered in chapter 9.

In the course of our work on the JPubE paper, Emmanuel and I uncovered that in the United States, given the values taken by the sufficient statistics, our formula for the efficient unemployment rate could be simplified to $u^* = \sqrt{uv}$. The value u^* also measures the full-employment rate of unemployment (FERU), because legal texts define full employment as the amount of employment maximizing social welfare.[9] These results were published in the *Brookings Papers on Economic Activity* in 2024 (Michaillat and Saez 2024b), and are presented in chapter 12.

[9] The full-employment mandate for the US federal government and central bank is defined by the Employment Act of 1946, the Federal Reserve Reform Act of 1977, and the Full Employment and Balanced Growth Act of 1978 (US Congress 1946, 1977; Congress 1978).

Are inflation and slack linked?

Much of the work in the book was motivated by the vast amount of slack that appeared in the US economy after the Great Recession. The aftermath of the pandemic, by contrast, led to historical tightness in the US labor market. These tight conditions led to new questions.

The first question was whether the excessive tightness on the US labor market could explain the inflation surge that occurred at the same time. Until then Emmanuel and I had assumed a fixed-inflation norm in our models because inflation had remained exceedingly stable in the decades prior to the pandemic. But the pandemic inflation pushed us to relax this assumption and study how slack and inflation could be connected.

We therefore developed a new theory of the Phillips (1958) curve connecting inflation to the unemployment gap (Michaillat and Saez 2024a). The Phillips curve emerged from directed search, as in Moen (1997), combined with a quadratic price-adjustment cost, as in Rotemberg (1982). The price-adjustment cost was required because without it, the Phillips curve would be degenerate: vertical, keeping unemployment at a fixed level, instead of downward-sloping. The slackish Phillips curve ensures that the divine coincidence described by Blanchard and Gali (2007) appears: inflation is on target when the economy is at full employment—in line with the evidence unearthed by Benigno and Eggertsson (2023). This work is the basis for chapter 14.

Does immigration affect slack?

A second question that arose after the pandemic is whether other policies besides cutting aggregate demand could help alleviate labor market tightness—thus helping firms fill vacant jobs and possibly dampening inflation. Naturally, another policy that could help was to expand labor supply via immigration. I therefore started to explore the effects of migration in a slackish labor market model, and confirmed that immigration reduces labor market tightness (Michaillat 2023). This work is discussed in chapter 11.

Has a new recession started?

Finally, after peaking in 2022, the US labor market cooled fairly rapidly in 2023–2025. This naturally led to the question: Has a new recession started? In the JPubE and BPEA papers, the combination of vacancy and unemployment data allowed us to compute the FERU and unemployment gap in real time. But it turns out that these data are useful to detect recessions in real time too—because recessions feature not only an increase in unemployment but also a decline in job vacancies as the economy moves down the Beveridge (1944) curve. By combining data on unemployment and job vacancies, Emmanuel and I constructed a recession rule (dubbed Michez rule by the *Financial Times*) that detects recessions faster and more robustly than the popular Sahm (2019) rule. It is even possible to do better than the Sahm and Michez rules by optimizing how the recession rules are constructed—by optimally filtering labor market data and selecting the detection threshold. The Michez rule was published in the *Oxford Bulletin of Economics and Statistics* in 2025 (Michaillat and Saez 2025); the optimal rules were described in Michaillat (2025); both papers are the foundation for chapter 18.

What is new in this book?

Together, the papers described above form the foundation of the book. Readers familiar with them will recognize various elements of the book. However, the book aims to be more than a collection of papers. It provides a unified and systematic treatment of the theory of slack, with consistent notation, concepts, and methodology. It also provides substantial evidence about slack—coming from diverse sources and spanning nearly one hundred years—to motivate and discipline the theory. Finally, the book places the model in context by clarifying its similarities and differences with leading business cycle models.

Many ingredients in the book's theory are not new: the matching function comes from the DMP literature; price and wage rigidities are ubiquitous in the Keynesian literature; wealth in the utility function has appeared before; sufficient statistics are commonplace in public economics; and so on. The novelty lies in how the book curates and combines these ingredients to offer a slack-centered theory of how economies operate and fluctuate, and to provide slack-based tools to monitor and respond to the cyclical inefficiencies that markets inevitably generate.

Bibliography

Akerlof, George A. 1980. "A Theory of Social Custom, of Which Unemployment May Be One Consequence." *Quarterly Journal of Economics* 94 (4): 749–775. https://doi.org/10.2307/1885667.

Akerlof, George A. 1982. "Labor Contracts as Partial Gift Exchange." *Quarterly Journal of Economics* 97 (4): 543–569. https://doi.org/10.2307/1885099.

Akerlof, George A. 1984. "Gift Exchange and Efficiency-Wage Theory: Four Views." *American Economic Review* 74 (2): 79–83. http://www.jstor.org/stable/1816334.

Akerlof, George A. and Janet L. Yellen. 1990. "The Fair Wage-Effort Hypothesis and Unemployment." *Quarterly Journal of Economics* 105 (2): 255–283. https://doi.org/10.2307/2937787.

Albrecht, James. 2011. "Search Theory: The 2010 Nobel Memorial Prize in Economic Sciences." *Scandinavian Journal of Economics* 113 (2): 237–259. https://doi.org/10.1111/j.1467-9442.2011.01658.x.

Arseneau, David M. and Sanjay K. Chugh. 2007. "Bargaining, Fairness, and Price Rigidity in a DSGE Environment." Board of Governors of the Federal Reserve System International Finance Discussion Paper 900. https://perma.cc/59RH-EAZD.

Asher, Martin A., Robert H. Defina, and Kishor Thanawala. 1993. "The Misery Index: Only Part of the Story." *Challenge* 36 (2): 58–62. https://doi.org/10.1080/05775132.1993.11471656.

Auerbach, Alan J. and Yuriy Gorodnichenko. 2012. "Measuring the Output Responses to Fiscal Policy." *American Economic Journal: Economic Policy* 4 (2): 1–27. https://doi.org/10.1257/pol.4.2.1.

Bai, Yan, Jose-Victor Rios-Rull, and Kjetil Storesletten. 2026. "Demand Shocks as Technology Shocks." *Review of Economic Studies* 93 (2): 798–832. https://doi.org/10.1093/restud/rdaf045.

Baily, Martin N. 1978. "Some Aspects of Optimal Unemployment Insurance." *Journal of Public Economics* 10 (3): 379–402. https://doi.org/10.1016/0047-2727(78)90053-1.

Bakshi, Gurdip S. and Zhiwu Chen. 1996. "The Spirit of Capitalism and Stock-Market Prices." *American Economic Review* 86 (1): 133–157. https://www.jstor.org/stable/2118259.

Barro, Robert J. 2025. "The Old Keynesian Model." NBER Working Paper 33850. https://doi.org/10.3386/w33850.

Barro, Robert J. and Herschel I. Grossman. 1971. "A General Disequilibrium Model of Income and Employment." *American Economic Review* 61 (1): 82–93. http://www.jstor.org/stable/1910543.

Benassy, Jean-Pascal. 1993. "Nonclearing Markets: Microeconomic Concepts and Macroeconomic Applications." *Journal of Economic Literature* 31 (2): 732–761. http://www.jstor.org/stable/2728514.

Benigno, Pierpaolo and Gauti B. Eggertsson. 2023. "It's Baaack: The Surge in Inflation in the 2020s and the Return of the Non-Linear Phillips Curve." NBER Working Paper 31197. https://doi.org/10.3386/w31197.

Beveridge, William H. 1944. *Full Employment in a Free Society*. London: George Allen & Unwin.

Blanchard, Olivier J. and Jordi Gali. 2007. "Real Wage Rigidities and the New Keynesian Model." *Journal of Money, Credit and Banking* 39 (1): 35–65. https://doi.org/10.1111/j.1538-4616.2007.00015.x.

Blanchard, Olivier J. and Jordi Gali. 2010. "Labor Markets and Monetary Policy: A New Keynesian Model with Unemployment." *American Economic Journal: Macroeconomics* 2 (2): 1–30. https://doi.org/10.1257/mac.2.2.1.

Boldrin, Michele, Nobuhiro Kiyotaki, and Randall Wright. 1993. "A Dynamic Equilibrium Model of Search, Production, and Exchange." *Journal of Economic Dynamics and Control* 17 (5–6): 723–758. https://doi.org/10.1016/0165-1889(93)90012-H.

Carroll, Christopher D. 2000. "Why Do the Rich Save So Much?" In *Does Atlas Shrug? The Economic Consequences of Taxing the Rich*, edited by Joel B. Slemrod. Cambridge, MA: Harvard University Press.

Chetty, Raj. 2006. "A General Formula for the Optimal Level of Social Insurance." *Journal of Public Economics* 90 (10-11): 1879–1901. https://doi.org/10.1016/j.jpubeco.2006.01.004.

Chetty, Raj. 2009. "Sufficient Statistics for Welfare Analysis: A Bridge Between Structural and Reduced-Form Methods." *Annual Review of Economics* 1: 451–488. https://doi.org/10.1146/annurev.economics.050708.142910.

Congress. 1978. "Full Employment and Balanced Growth Act of 1978." FRASER, Federal Reserve Bank of St. Louis. https://fraser.stlouisfed.org/title/1034.

Diamond, Peter. 2011. "Unemployment, Vacancies, Wages." *American Economic Review* 101 (4): 1045–1072. https://doi.org/10.1257/aer.101.4.1045.

Diamond, Peter and Drew Fudenberg. 1989. "Rational Expectations Business Cycles in Search Equilibrium." *Journal of Political Economy* 97 (3): 606–619. https://doi.org/10.1086/261618.

Diamond, Peter and Joel Yellin. 1990. "Inventories and Money Holdings in a Search Economy." *Econometrica* 58 (4): 929–950. https://doi.org/10.2307/2938356.

Diamond, Peter A. 1982. "Aggregate Demand Management in Search Equilibrium." *Journal of Political Economy* 90 (5): 881–894. https://doi.org/10.1086/261099.

Diamond, Peter A. 1984. "Money in Search Equilibrium." *Econometrica* 52 (1): 1–20. https://doi.org/10.2307/1911458.

Eyster, Erik, Kristof Madarasz, and Pascal Michaillat. 2021. "Pricing under Fairness Concerns." *Journal of the European Economic Association* 19 (3): 1853–1898. https://doi.org/10.1093/jeea/jvaa041.

Gourio, Francois and Leena Rudanko. 2014. "Customer Capital." *Review of Economic Studies* 81 (3): 1102–1136. https://doi.org/10.1093/restud/rdu007.

Hall, Robert E. 2005. "Employment Fluctuations with Equilibrium Wage Stickiness." *American Economic Review* 95 (1): 50–65. https://doi.org/10.1257/0002828053828482.

Hall, Robert E. 2007. "Equilibrium Sticky Prices." https://perma.cc/XF9C-493X.

Katz, Lawrence F. 1986. "Efficiency Wage Theories: A Partial Evaluation." *NBER Macroeconomics Annual* 1: 235–276. https://doi.org/10.1086/654025.

Keynes, John Maynard. 1936. *The General Theory of Employment, Interest and Money*. New York: Macmillan.

Kleven, Henrik J. 2021. "Sufficient Statistics Revisited." *Annual Review of Economics* 13: 515–538. https://doi.org/10.1146/annurev-economics-060220-023547.

Kumhof, Michael, Romain Ranciere, and Pablo Winant. 2015. "Inequality, Leverage, and Crises." *American Economic Review* 105 (3): 1217–1245. https://doi.org/10.1257/aer.20110683.

Kurz, Mordecai. 1968. "Optimal Economic Growth and Wealth Effects." *International Economic Review* 9 (3): 348–357. https://doi.org/10.2307/2556231.

Landais, Camille, Pascal Michaillat, and Emmanuel Saez. 2018a. "A Macroeconomic Approach to Optimal Unemployment Insurance: Applications." *American Economic Journal: Economic Policy* 10 (2): 182–216. https://doi.org/10.1257/pol.20160462.

Landais, Camille, Pascal Michaillat, and Emmanuel Saez. 2018b. "A Macroeconomic Approach to Optimal Unemployment Insurance: Theory." *American Economic Journal: Economic Policy* 10 (2): 152–181. https://doi.org/10.1257/pol.20150088.

Lehmann, Etienne and Bruno Van der Linden. 2010. "Search Frictions on Product and Labor Markets: Money in the Matching Function." *Macroeconomic Dynamics* 14 (1): 56–92. https://doi.org/10.1017/S1365100509080316.

Matha, Thomas Y. and Olivier Pierrard. 2011. "Search in the Product Market and the Real Business Cycle." *Journal of Economic Dynamics & Control* 35 (8): 1172–1191. https://doi.org/10.1016/j.jedc.2011.03.001.

Michaillat, Pascal. 2012. "Do Matching Frictions Explain Unemployment? Not in Bad Times." *American Economic Review* 102 (4): 1721–1750. https://doi.org/10.1257/aer.102.4.1721.

Michaillat, Pascal. 2014. "A Theory of Countercyclical Government Multiplier." *American Economic Journal: Macroeconomics* 6 (1): 190–217. https://doi.org/10.1257/mac.6.1.190.

Michaillat, Pascal. 2023. "Modeling Migration-Induced Unemployment." arXiv:2303.13319. https://doi.org/10.48550/arXiv.2303.13319.

Michaillat, Pascal. 2025. "Recession Detection Using Classifiers on the Anticipation-Precision Frontier." arXiv:2506.09664. https://doi.org/10.48550/arXiv.2506.09664.

Michaillat, Pascal and Emmanuel Saez. 2015. "Aggregate Demand, Idle Time, and Unemployment." *Quarterly Journal of Economics* 130 (2): 507–569. https://doi.org/10.1093/qje/qjv006.

Michaillat, Pascal and Emmanuel Saez. 2019. "Optimal Public Expenditure with Inefficient Unemployment." *Review of Economic Studies* 86 (3): 1301–1331. https://doi.org/10.1093/restud/rdy030.

Michaillat, Pascal and Emmanuel Saez. 2021a. "Beveridgean Unemployment Gap." *Journal of Public Economics Plus* 2: 100009. https://doi.org/10.1016/j.pubecp.2021.100009.

Michaillat, Pascal and Emmanuel Saez. 2021b. "Resolving New Keynesian Anomalies with Wealth in the Utility Function." *Review of Economics and Statistics* 103 (2): 197–215. https://doi.org/10.1162/rest_a_00893.

Michaillat, Pascal and Emmanuel Saez. 2022. "An Economical Business-Cycle Model." *Oxford Economic Papers* 74 (2): 382–411. https://doi.org/10.1093/oep/gpab021.

Michaillat, Pascal and Emmanuel Saez. 2024a. "Beveridgean Phillips Curve." arXiv:2401.12475. https://doi.org/10.48550/arXiv.2401.12475.

Michaillat, Pascal and Emmanuel Saez. 2024b. "$u^* = \sqrt{uv}$: The Full-Employment Rate of Unemployment in the United States." *Brookings Papers on Economic Activity* 55 (2): 323–390. https://doi.org/10.1353/eca.2024.a964373.

Michaillat, Pascal and Emmanuel Saez. 2025. "Has the Recession Started?" *Oxford Bulletin of Economics and Statistics* 87 (6): 1047–1058. https://doi.org/10.1111/obes.12685.

Moen, Espen R. 1997. "Competitive Search Equilibrium." *Journal of Political Economy* 105 (2): 385–411. https://doi.org/10.1086/262077.

Mortensen, Dale T. 2011. "Markets with Search Friction and the DMP Model." *American Economic Review* 101 (4): 1073–1091. https://doi.org/10.1257/aer.101.4.1073.

Okun, Arthur M. 1981. *Prices and Quantities: A Macroeconomic Analysis*. Washington, DC: Brookings Institution.

Petrosky-Nadeau, Nicolas and Etienne Wasmer. 2015. "Macroeconomic Dynamics in a Model of Goods, Labor and Credit Market Frictions." *Journal of Monetary Economics* 72: 97–113. https://doi.org/10.1016/j.jmoneco.2015.01.006.

Phillips, A. W. 1958. "The Relation Between Unemployment and the Rate of Change of Money Wage Rates in the United Kingdom, 1861–1957." *Economica* 25 (100): 283–299. https://doi.org/10.1111/j.1468-0335.1958.tb00003.x.

Piketty, Thomas. 2011. "On the Long-Run Evolution of Inheritance: France 1820–2050." *Quarterly Journal of Economics* 126 (3): 1071–1131. https://doi.org/10.1093/qje/qjr020.

Pissarides, Christopher A. 1979. "Job Matchings with State Employment Agencies and Random Search." *Economic Journal* 89 (356): 818–833. https://doi.org/10.2307/2231501.

Pissarides, Christopher A. 1984. "Efficient Job Rejection." *Economic Journal* 94 (Supplement): 97–108. https://doi.org/10.2307/2232658.

Pissarides, Christopher A. 1985. "Short-Run Equilibrium Dynamics of Unemployment, Vacancies, and Real Wages." *American Economic Review* 75 (4): 676–690. https://www.jstor.org/stable/1821347.

Pissarides, Christopher A. 1986. "Unemployment and Vacancies in Britain." *Economic Policy* 1 (3): 500–559. https://doi.org/10.2307/1344583.

Pissarides, Christopher A. 2000. *Equilibrium Unemployment Theory*. 2nd ed. Cambridge, MA: MIT Press.

Pissarides, Christopher A. 2011. "Equilibrium in the Labor Market with Search Frictions." *American Economic Review* 101 (4): 1092–1105. https://doi.org/10.1257/aer.101.4.1092.

Rotemberg, Julio J. 1982. "Monopolistic Price Adjustment and Aggregate Output." *Review of Economic Studies* 49 (4): 517–531. https://doi.org/10.2307/2297284.

Saez, Emmanuel. 2001. "Using Elasticities to Derive Optimal Income Tax Rates." *Review of Economic Studies* 68 (1): 205–229. https://doi.org/10.1111/1467-937X.00166.

Sahm, Claudia. 2019. "Direct Stimulus Payments to Individuals." In *Recession Ready: Fiscal Policies to Stabilize the American Economy*, edited by Heather Boushey, Ryan Nunn, and Jay Shambaugh, chap. 3. Washington: Brookings Institution. `https://www.brookings.edu/articles/direct-stimulus-payments-to-individuals/`.

Samuelson, Paul A. 1954. "The Pure Theory of Public Expenditure." *Review of Economics and Statistics* 36 (4): 387–389. `https://doi.org/10.2307/1925895`.

Samuelson, Paul A. 1955. "Diagrammatic Exposition of a Theory of Public Expenditure." *Review of Economics and Statistics* 37 (4): 350–356. `https://doi.org/10.2307/1925849`.

Shimer, Robert. 2005. "The Cyclical Behavior of Equilibrium Unemployment and Vacancies." *American Economic Review* 95 (1): 25–49. `https://doi.org/10.1257/0002828053828572`.

Tobin, James. 1992. "An Old Keynesian Counterattacks." *Eastern Economic Journal* 18 (4): 387–400. `https://www.jstor.org/stable/40325470`.

Tobin, James. 1993. "Price Flexibility and Output Stability: An Old Keynesian View." *Journal of Economic Perspectives* 7 (1): 45–65. `https://doi.org/10.1257/jep.7.1.45`.

US Congress. 1946. "Employment Act of 1946." FRASER, Federal Reserve Bank of St. Louis. `https://fraser.stlouisfed.org/title/1099`.

US Congress. 1977. "Federal Reserve Reform Act of 1977." FRASER, Federal Reserve Bank of St. Louis. `https://fraser.stlouisfed.org/title/1040`.

Walsh, Carl E. 1986. "The Macroeconomic Effects of Microeconomic Trading Uncertainty." `https://perma.cc/S39D-NYS8`.

Yellen, Janet L. 1984. "Efficiency Wage Models of Unemployment." *American Economic Review* 74 (2): 200–205. `https://www.jstor.org/stable/1816355`.

Zou, Heng-fu. 1994. "The Spirit of Capitalism and Long-Run Growth." *European Journal of Political Economy* 10: 279–293. `https://doi.org/10.1016/0176-2680(94)90020-5`.